highways
england

Our ref: A419
Your ref: S/OUT/13/1555

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8 April 2015

Swindon Borough Council
Wat Tyler House
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**For the attention of Miss Sam Howell**

Dear Sam

**M4/A419: PROPOSED DEVELOPMENT – EASTERN VILLAGES (ROWBOROUGH AND SOUTH MARSTON)**

I refer to the above application in relation to which Highways England has had correspondence and discussion with you as your client's appointed transport consultants.

We have now had an opportunity to review your latest transport assessment and travel plan and for ease of reference our comments are included in the attached technical notes.

Please contact me if you wish to discuss further.

Yours sincerely

Rachel Sandy
Asset Manager
NDD South West – Growth & Improvement Team
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**Review of Brookbanks Transport Assessment Rev3 Jan/Feb 2015**

Comments provided under headings from December 2013 JMP technical note

**General Transport Assessment Comments**

The comparative masterplan between that tested by Halcrow and JMP on behalf of the developer and that shown in the TA has not been provided.

**Chapter 3 - National and Local Policy Background**

TA has been updated to acknowledge HA Circular 02/2013 as requested.

**Chapter 4 – Existing Transport Conditions**

A more detailed plan of the junctions on the SRN and locations of the individual accidents, together with descriptions would still be beneficial, as it is not clear if the accidents are a result of collisions between circulatory and slip road traffic or because of queuing on the approaches to the junction.

An assessment of the accident data has been carried out for the A419 to the north of the Whitehart junction to the Blunsden junction (inclusive) within the TA. The comment above would still apply and further information would help understand the linear pattern of collisions to the south of the Hyde Road junction and either side of the Blunsden junction (i.e. is this due to slow moving or stationary traffic on the approaches) for instance.

The TA considers that the higher than average occurrence of rear end shunts at the White Hart roundabout is principally concerning drivers not slowing sufficiently on the approaches and/or judging correctly when it is safe to proceed. The TA suggests that this would be improved with the introduction of traffic signals because this is a more formalised junction control and drivers' decisions are made easier and there is less room for misjudgement to occur. However, traffic signals would still be associated with queuing traffic and may introduce a different type of safety issue (e.g. red light violations / overshoots) and this should be considered and mitigated as required (e.g. maximum visibility to signals / queuing traffic).

**Chapter 5 – Development Proposals**

A detailed phasing plan has not been provided, as previously requested. This would help determine appropriate triggers for the delivery of infrastructure to facilitate the development.

The public transport strategy has been expanded and contains more detail together with a proposed bespoke bus service to serve the site. However, further detail is requested to confirm how the bus service and the bus contribution is linked to the development build out, the forecast bus patronage and modal shift, to ensure this is a suitable measure to help minimise trips on the SRN. An assessment beyond the funding period is also recommended to help determine how viable and sustainable such a service would be in the medium to long term.

It is noted that there is no significant detail on the potential parking provision. It is appreciated that this is an outline application, but a clear strategy for the parking provision and how this will help minimise vehicle trips on the SRN, with reference to SBCs parking standards is strongly advised.

**Chapter 6 – Development Impact Appraisal**

A detailed phasing plan has not been provided, as previously requested. This would help understand the number of dwellings and associated trips forecast through the build out, which can be used to identify any gaps before proposed mitigation measures come on stream.

**Chapter 7 – Travel Plan**

Reviewed separately. To follow.

**Chapter 8 – Development Trip Generation**

A 2016 opening year assessment with the full application proposal is now included within the TA, based on the PARAMICS modelling. It includes highway mitigation at the Gablecross Roundabout (conversion to a signalised crossroads), and the partial signalisation of the White Hart Junction and the Merlin Way / A419 slip road roundabout in order to accommodate the proposed development. A 2026 year assessment is also included for a future year assessment, with the application site plus future development at Eastern Villages. The TA also expands on the previous version that relied principally on GEH statistics.

**Chapter 9 – Local Road Network Review – Journey Times**

The 2016 base plus development scenario, shows increases in journey times on the following SRN link compared to the base situation:

- Route 3 PM – A419 Northbound. M4 J15 to A420 Gable Cross Roundabout = 84secs

The 2016 base plus development scenario, shows notable decreases in average journey speeds on the following SRN links compared to the base situation:

- Route 3 PM – A419 Northbound. M4 J15 to A420 Gable Cross Roundabout = -10kph

The TA states that the increase in journey time is within the daily variation and would not be noticeable and is not a severe impact. In order to determine whether this would be classified as a severe impact a cumulative analysis taking into account the queue lengths is also required.

The journey times and speeds on Route 3 are markedly improved with the White Hart Interchange enhanced to a signalised roundabout or as a diamond interchange in the 2026 future year assessments. It would therefore be beneficial to understand the development build-out for the application site, plus phasing of the wider Eastern Villages if possible, so as to help determine when the appropriate mitigation would be best brought forward.

The average speed on the A420 eastbound from White Hart to east of Old Vicarage Lane is also shown to decrease by approximately six and seven seconds AM and PM respectively, as a result of the development in 2016. The journey times on the A420 eastbound from White Hart to east of Old Vicarage Lane is also shown to increase significantly by 83 seconds and 104 seconds in the AM and PM respectively, as a result of the development. This should be considered alongside the queue length analysis in order to determine whether there is the potential for queuing or slow moving traffic that could potentially back up to the White Hart Interchange.

**Chapter 10 – Link Flow Assessment**

The broad link flow assessment shows the A419 to be potentially associated with an increase in traffic of between approximately three and five percent in the peak hours in 2016, which on a congested network such as this, could impact on the safety and operation of the SRN. However, there are significant increases in traffic on the A420 between White Hart Interchange and Gablecross roundabout, which could have an impact on the A420 entry and exit arms and the White Hart circulatory carriageway. The detailed junction analysis at White Hart and Gablecross will help more accurately understand the traffic impact on the SRN.

The use of the theoretical capacity of the links is only considered to be a guide to assess development traffic impact. This assumes all carriageways are built to DMRB standards and there is no interaction between the different links, or other features such as pedestrian crossings that affect flow. DMRB TA79/99 states that the link capacity for different road types is a guide and may be used as a starting point in the assessment of the capacity of an existing urban road.

Notwithstanding the TA stating that the assessment of the two way link flow indicates that the majority of the links operate well within theoretical highway capacity limits, the actual increase in vehicle flow arising from the application site on each link in 2016 is significant.

**Chapter 11 Junction Assessment**

The junction assessment as currently presented provides some details on the increase/ decrease in queue lengths but this needs to be considered within the context of whether these block back to the mainline. Despite previous advice we are disappointed that this contextual information has not been provided. It is therefore difficult to understand within the body of the TA as to whether the queues could be considered material. This is requested of the applicant.

M4 Junction 15

The modelling results indicate that queues on the slip-roads will not increase significantly as a result of the development, but there could be an increase in queuing on the A419 approach of up to around 30 percent (21 metres) in 2016. However, this is not likely to impede traffic joining the A419 from the Commonhead roundabout and therefore is not likely to be significant.

Rat Trap Roundabout

The increase in queuing at the junction is not forecast to be significant in 2016 with development, but the potential queuing on the western approach (i.e. A419 northbound off-slip) in 2026 with development is high and further evidence to confirm that the queuing will not affect the safe and efficient operation of the A419 mainline will be required, or mitigation measures proposed or contributed to as part of the IDP.

Commonhead Junction

This development does not appear to significantly impact upon Commonhead Junction, but significant increases in queue length is forecast in 2026 with the full Eastern Villages expansion. This

would need to be considered in the context of potential impact on the A419 mainline and mitigation through wider IPD improvement measures.

Blunsdon Bypass

This development does not appear to significantly impact upon Blunsdon Bypass Junctions and the impact that could have on the SRN.

White Hart Interchange

The TA modelling is showing significant increases in queuing on the A420 eastern approach and the Merlin Way approach to the roundabout in 2016 with the proposed development. In 2026 with full Eastern Villages development there will be significant increases in queue length on Oxford Road approach and the A419 southbound off-slip.

The timing of the mitigation measures at this junction is therefore critical as the application site and wider development is built out. Further evidence that the queuing will not affect the safe and efficient operation of the A419 mainline will be required.

Merlin Way/A419 Slip Road

Clarification is required on the content of Figure 11ae, as no 2016 assessment appear to be presented. From the results presented, the development does not appear to significantly impact upon the queue lengths on the A419 slip road, but this should be confirmed in terms of the impact any queuing could have on the SRN mainline flows.

Gablecross Roundabout

The results of the 2016 assessment indicate that the queue is predicted to increase on all the arms. The queue on the A420 approach from White Hart Interchange is forecast to increase by 113 metres with the introduction of development traffic and this would take the average queue length to 169 metres. On this basis, the queue would not reach as far back to the White Hart Interchage, but given previous modelling results presented in the TA, there could be slow moving traffic on the A420 affecting the rate of dispersal from the White Hart Interchange. In 2026 the queues forecast on the A420 eastbound could potentially queue back to the White Hart Interchange. The linking of the signals at the two junctions would therefore be important to ensure there is no impact from queuing or slow moving traffic on the A420 approaching Gablecross. Further examination of the interaction between the two junctions is recommended, particularly given SBCs comments on the design of the Gablecross junction.

**Conclusions**

The additional analysis and assessments within the revised TA have been useful and have helped resolve or clarify some issues. However, there are a number of issues that will need to be addressed and information provided to satisfy the HA, particularly in respect of the public transport strategy, accident analysis and the traffic impact at and between White Hart Interchange and Gable Cross Roundabout through a more detailed queue analysis.

# JMP review of Brookbanks Travel Plan Rev3 Dated 11/04/14

## General Comments

1. This Technical Note relates to the Travel Plan Framework (TPF) Revision 3 produced by Brookbanks Consultants Limited dated 11/04/14. JMP reviewed the previous version Travel Plan (dated 30/09/13) and prepared a response dated 04/12/13.

2. The comments are provided under headings from the December 2013 JMP technical note, where appropraite.

3. This version of the TPF has addressed a number of comments raised by JMP, but there are still outstanding matters that are considered to need addressing, as set out below. Subject to these points being addressed, overall it is recommended that the TPF document is of good enough quality to be approved by The Agency.

## Section 2: Travel Plan Principles

4. Paragraph 2.2 refers to a Sustainable Transport Strategy for Swindon EV. It is still not clear whether this is a separate document or an overarching strategy or aspiration. If it is a separate document this should be appended to the TPF and the TPF should highlight where it accords with this Strategy.

## Section 5: Travel Plan Objectives and Targets

5. Paragraphs 5.5 and 5.6 specify the appointment of the Site Wide Travel Plan Coordinator (SWTPC) and also the individual travel plan coordinators for the subsequent developments on the site, which is helpful. However, clarification is required that a SWTPC will be appointed for this development site and will be position 3 months prior to occupation of the site, as previously recommended.

6. This section has identified a broad target for reducing Single Occupancy Vehicle (SOV) trips by 10% over 10 years. However, there is no detailed timetable of when this starts / finishes and interim targets should also be set to reflect the monitoring periods and development build out.

7. There is no indication of what modes will increase to accommodate this decrease in SOV trips (paragraph 5.11). However, it should be possible to identify where it is thought the mode shift is going to come from, based on the targeted measures and initiatives. This should therefore be provided.

8. Table 5.1 should now identify the SOV target set and the appropriate targets for walking, cycling and using public transport.

## Section 6: Eastern Villages Travel Plan Implementation Strategy

9. Clarification is needed for Paragraph 6.3, which includes reference to a Travel Plan Steering Group. It is assumed this is the Eastern Villages Steering Group (EVSG) referred to in paragraph 1.14, but this needs confirming and should be consistent throughout the report.

10. The benefits of the employment Transport Management Group (TMG) are identified, but the benefits of the EVSG also need to be identified; and how the developers and employers will

be encouraged / required to sign up to the groups should also be explained (e.g. S106 or other agreements). Detail on how the groups are funded and how the monies will be secured is necessary.

11. It is considered that more detail is required on the SWTPC and in particular the associated costs and indication of the time, resources, management support and finance required. This would help identify at an early stage the resources that will be required to ensure the successful delivery of the travel planning at the site.

## Section 7: Site Wide Measures

12. Paragraph 7.5 The Bicycle User Group section refers to an annual £250 budget funded by the TMG. Clarification is sought as to whether this covers both the residents and employees and which group it will be funded through - it is currently suggested it is funded by the SMG for the employment elements.

13. Paragraph 7.5 The Car Share Database section refers to the TMG paying the license subscription charge. Clarification is sought as to whether this is funded through the employment or residential element of the development, as it would benefit both sets of people.

14. The action plan table now details the car club being implemented upon the 250th occupation, but this is not consistent with the preceding text. Further detail is sought on the car club funding and the membership / take-up of such a service and how and when it could become self-funding. It may still be possible to establish a community based car club from first occupation to provide this option to residents and build up membership. This could ultimately tie in with the wider Eastern Villages developments.

15. There is still no mention of electric charge points or pool bikes being provided on the site, which are measures that should also be considered for a development of this size and nature.

16. It would be helpful to set out an approximate value of the bus taster tickets. This would also help with establishing the Travel Plan funding required for the site.

## Section 8: Monitoring

17. Currently the monitoring surveys are set for completion of each 800 units, meaning three sets of surveys (800, 1600, 2,380) plus a further two years of annual monitoring. However, this could be too long between surveys, depending on the build-out. Ideally, the best way to monitor progress is by carrying out annual 'snapshot' surveys (e.g. ATCs), with full travel surveys carried out every 2 to 3 years. The monitoring section therefore needs to be expanded and clarification on the monitoring schedule is required and how this is tied to the site build-out, occupation projections and the targets.

18. It is requested that more evidence and detail of the monitoring methodology is provided. Additional surveys/diaries/interviews with residents on the site should also be collected in addition to quantitative data through ATCs.

19. Paragraph 8.6 refers to a failsafe fund provided by the developer to implement remedial measures, if the TPF does not achieve its targets. Confirmation on how this will be secured (e.g. S106) is required together with confirmation that the remedial measures will be agreed through discussions with Swindon Borough Council and the HA, as required.